

BORYEONG, Korea, Republic ofWMO: **471500**Lat: **36.3272N**Lon: **126.5575E**Elev: **17**StdP: **101.12**Time Zone: **9.00 (E09)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-7.3	-16.0	0.9	-5.9	-14.3	1.1	-4.9	5.6	3.1	5.2	2.4	2.1	20	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	31.8	26.6	30.5	26.0	29.2	25.2	27.8	30.1	27.0	29.0	26.2	28.2	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.1	29.2	26.5	22.0	28.4	25.7	21.0	27.7	89.4	30.5	85.7	29.0	82.0	28.2	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	5.7	4.8	-10.4	33.7	2.5	1.3	-12.1	34.7	-13.6	35.4	-15.0	36.2	-16.7	37.1		
(5)			-11.3	27.9	2.1	1.5	-12.9	29.0	-14.1	29.9	-15.3	30.7	-16.8	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-0.7	1.5	5.6	10.9	16.8	21.5	25.2	26.2	21.7	15.6	9.0	1.9
	DBStd	9.70	3.70	3.73	3.43	3.35	2.81	1.83	2.00	2.29	2.25	3.06	4.16	4.05
	HDD10.0	1068	332	239	145	29	0	0	0	0	0	2	68	252
	HDD18.3	2627	590	472	395	223	63	2	0	2	92	279	509	
	CDD10.0	2164	0	1	8	57	212	345	472	503	349	177	39	1
	CDD18.3	682	0	0	0	1	16	97	213	244	102	8	1	0
	CDH23.3	4813	0	0	0	4	62	346	1628	2302	450	21	0	0
	CDH26.7	1148	0	0	0	0	3	24	348	719	52	2	0	0
Wind	WSAvg	1.8	1.6	1.7	1.9	1.9	1.9	1.6	2.3	1.8	1.5	1.5	1.8	1.8
Precipitation	PrecAvg	1208	28	32	47	87	89	134	265	253	132	47	58	34
	PrecMax	1897	71	89	96	186	238	328	631	657	282	123	164	69
	PrecMin	725	5	2	2	16	10	5	81	93	4	4	12	8
	PrecStd	306	20	26	26	49	53	91	138	135	88	31	38	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	17.0	18.5	23.1	27.0	28.6	32.7	34.1	29.7	25.7	21.1	15.0
		MCWB	8.4	13.3	14.0	13.9	17.3	20.7	26.1	27.0	23.0	19.3	16.2	11.3
	2%	DB	8.6	12.2	15.8	20.0	25.0	27.3	30.9	32.5	28.1	23.5	18.8	11.5
		MCWB	6.2	9.7	12.2	13.6	17.5	20.8	26.6	26.8	22.4	18.5	15.0	8.6
	5%	DB	6.5	9.1	13.8	18.5	23.6	26.2	29.6	31.4	26.9	22.1	16.8	9.6
		MCWB	4.0	5.7	10.2	12.8	16.9	20.6	25.9	26.4	22.0	17.2	13.4	7.1
	10%	DB	5.0	7.2	11.7	17.0	22.2	25.3	28.5	30.2	25.8	20.8	15.4	7.9
		MCWB	2.6	4.2	8.1	12.1	16.3	20.5	25.4	25.9	21.3	16.2	12.5	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	13.7	15.0	17.6	22.1	24.2	28.7	29.3	25.5	22.0	17.5	12.4
		MCDB	10.7	16.3	17.1	19.1	23.0	25.9	30.4	31.6	28.0	24.7	20.1	14.1
	2%	WB	6.4	9.7	13.2	15.4	19.3	23.3	27.7	28.2	24.2	20.2	16.1	9.3
		MCDB	8.5	11.9	15.0	18.5	21.9	24.9	29.4	30.9	26.6	22.4	17.7	11.1
	5%	WB	4.4	6.4	10.9	14.3	18.2	22.4	27.0	27.4	23.4	18.7	14.3	7.2
		MCDB	6.2	8.5	13.5	17.2	21.5	24.4	28.5	29.8	25.6	20.4	16.2	9.0
	10%	WB	2.8	4.6	8.7	13.0	17.3	21.5	26.2	26.6	22.6	17.2	12.9	5.6
		MCDB	4.6	6.8	11.1	15.9	20.7	23.9	27.7	28.9	24.8	19.4	15.0	7.7
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	8.5	9.0	8.6	7.0	5.1	5.9	7.3	8.6	7.4	6.9
		MCDBR	8.2	10.1	10.4	11.0	10.3	8.2	6.4	7.5	8.3	9.3	8.1	8.1
		MCWBR	6.2	7.7	7.3	6.2	5.0	3.7	3.2	3.3	4.0	5.1	5.5	6.7
	5% WB	MCDBR	7.7	9.5	9.1	9.0	8.3	6.1	5.5	6.6	6.2	7.3	7.1	7.6
		MCWBR	6.5	8.0	7.2	5.9	5.3	3.7	3.5	3.5	3.5	5.3	5.9	7.0
Clear-Sky Solar Irradiance	taub	0.457	0.499	0.577	0.602	0.622	0.694	0.646	0.581	0.510	0.497	0.481	0.457	
	taud	1.911	1.844	1.703	1.694	1.706	1.632	1.783	1.940	2.055	1.952	1.937	1.928	
	Ebn at Noon	684	710	694	708	702	652	681	718	747	705	656	651	
	Edh at Noon	147	177	223	237	237	256	218	183	155	157	143	137	
All-Sky Solar Radiation	RadAvg	2.37	3.22	4.40	5.12	5.63	5.17	4.40	4.59	4.22	3.66	2.47	2.07	
	RadStd	0.16	0.29	0.43	0.40	0.56	0.52	0.64	0.67	0.43	0.29	0.28	0.16	